

Systems Analysis & Control: Project II

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Linearized Model:

The given transfer functions below are linearized about $\dot{y}_0 = -100\text{m/s}$, and $T_0 = 0.7 \times T_{\max}$

$$G_\theta(s) = \frac{\Theta(s)}{U(s)} = \frac{1.61(s + 0.833)}{(s + 0.213)(s^2 + 0.206s + 3.41)} \quad (1)$$

$$G_x(s) = \frac{X(s)}{U(s)} = \frac{-0.337(s^2 + 5.20)}{s(s + 0.213)(s^2 + 0.206s + 3.41)} \quad (2)$$

Task:

Select a controller, $u(t)$, in order to control $x(t) \rightarrow 0$ & $\theta(t) \rightarrow 0$ (landing upright on landing pad).

Using a rough block diagram below:

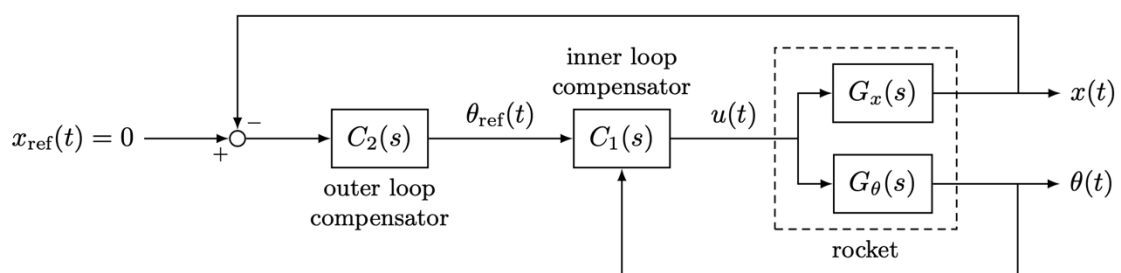


Figure 1: Block diagram containing control and plant to output $x(t)$ and $\theta(t)$

(a) Designing the inner loop compensator $C_1(s)$:

From Figure 1, the first step involved designing the inner loop compensator $C_1(s)$. The equation to model the controller is given below.

$$u(t) = K_p(\theta_{\text{ref}}(t) - \theta(t)) - K_d\dot{\theta}(t) \quad (3)$$

Where K_p is the proportional gain, K_d is the derivative gain, and $\theta_{\text{ref}}(t)$ is the reference angle with respect to time. The goals for designing $C_1(s)$ are such that:

- Maximum overshoot is less than 10%
- Settling time is less than 2.5 seconds
- $K_p < 20$, which will ensure the commanded $u(t)$ is not too large.

The gain values can be determined using the root locus method. First, the damping ratio (ζ) is determined using the equation for settling time (t_s) and percent overshoot (M_p) given by:

$$t_s = \frac{4}{\zeta\omega_n} \quad \& \quad M_p = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} \quad (4)$$

Rearranged and using the predetermined criteria,

$$\zeta = \frac{-\ln(M_p)}{\sqrt{\pi^2 + \ln^2(M_p)}}, \quad \zeta = -\ln(0.1)/\sqrt{\pi^2 + \ln^2(0.1)} = \mathbf{0.591155} \quad (5)$$

The damping ratio can then be used to determine the angle (θ) for the feasibility region of the root locus diagram.

$$\cos \theta = \zeta, \quad \theta = \cos^{-1}(0.5912) = 53.76^\circ \quad (6)$$

The same can be done for the real parts of the root locus diagram from settling time requirement.

$$\zeta\omega_n = \frac{4}{t_s}, \quad \text{Re}(s) = -\zeta\omega_n = \frac{-4}{2.5} = -1.6 \quad (7)$$

Creating root locus diagram with the constraints $\theta < 53.76^\circ$ & $\text{Re}(s) < -1.6$ gives the admissible region shown Figure 2.

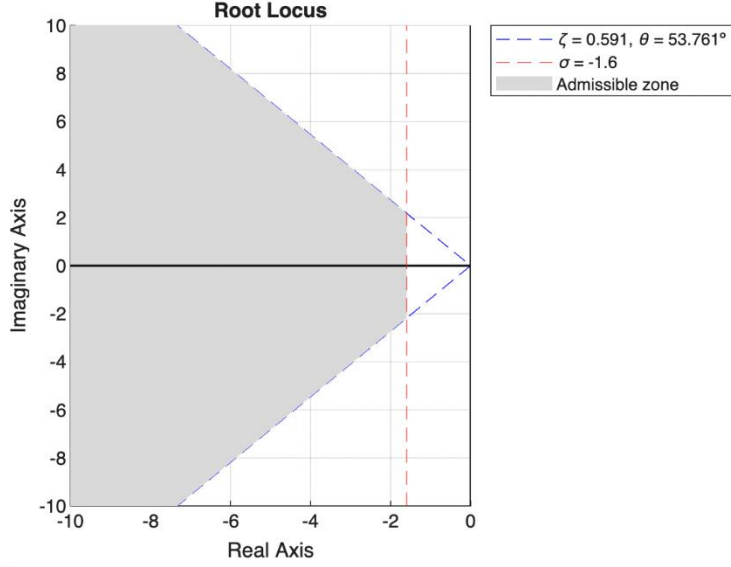


Figure 2: Root locus of feasible solutions based on the settling time and percent overshoot requirements

Next, the characteristic equation is used to determine the gain values.

$$G_{\theta}(s) = \frac{\Theta(s)}{U(s)} = \frac{1.61(s+0.833)}{(s+0.2133)(s^2+0.2065s+3.41)}, \quad \Theta(s) = G_{\theta}(s)U(s) \quad (8)$$

$$\Theta(s) = G_{\theta}(s)[K_p(\Theta_{ref}(s) - \Theta(s)) - K_d s \Theta(s)] \quad (9)$$

$$\Theta(s) + (K_p + K_d s)G_{\theta}(s)\Theta(s) = G_{\theta}(s)K_p\Theta_{ref}(s) \quad (10)$$

$$\Theta(s)[1 + (K_p + K_d s)G_{\theta}(s)] = G_{\theta}(s)K_p\Theta_{ref}(s) \quad (11)$$

$$\frac{\Theta(s)}{\Theta_{ref}(s)} = \frac{K_p G_{\theta}(s)}{1 + (K_p + K_d s)G_{\theta}(s)} \quad (12)$$

$$\begin{aligned} \frac{\Theta(s)}{\Theta_{ref}(s)} &= \frac{K_p[1.61(s + 0.833)]}{((s + 0.2133)(s^2 + 0.2065s + 3.41) + (K_p + K_d s)1.61(s + 0.833))} \\ &= \frac{K_p[1.61(s + 0.833)]}{s^3 + (0.419 + 1.61K_d)s^2 + (3.453878 + 1.34113K_d + 1.61K_p)s + (0.72633 + 1.34113K_p)} \end{aligned} \quad (13)$$

From the closed-loop transfer function, the characteristic equation is therefore,

$$\begin{aligned} s^3 + (0.419 + 1.61K_d)s^2 + (3.453878 + 1.34113K_d + 1.61K_p)s \\ + (0.72633 + 1.34113K_p) = 0 \end{aligned} \quad (15)$$

It can be seen that K_p and K_d influence different coefficients of the polynomial and therefore affect the closed-loop pole locations from Routh's Criteria. The proportional gain K_p appears in the coefficients of s^1 and s^0 and mainly increases the response speed so that the settling time requirement can be met. Since the specification also required $K_p < 20$, a value of $K_p = 19$ can be selected as it is close to the upper limit of the constraint but still gives a fast response. The derivative gain K_d affects the s^2 and s coefficients, so it mainly influences the damping of the system. Increasing K_p will increase damping, which reduces oscillation and overshoot. A value of $K_d = 4$ can be selected because it produces a sufficiently well-damped response, keeping the maximum overshoot below 10% while maintaining a settling time below 2.5 s.

Picking $K_p = 19$ and $K_d = 4$ for this controller is therefore suitable.

Giving,

$$C_1(s) = 19 + 4s$$

(16)

The controller was then used to create the closed loop transfer function in MATLAB (see Appendix) which was graphed as shown below.

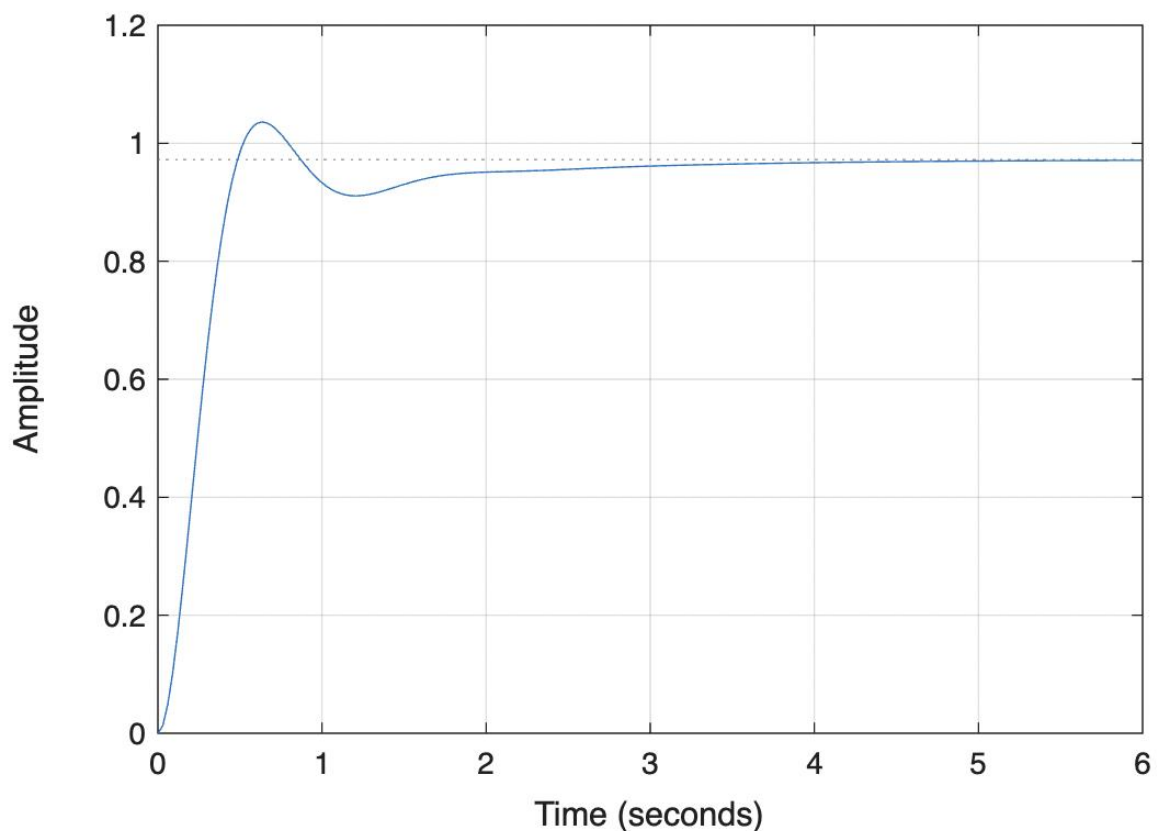


Figure 3: Closed loop transfer function graph of $G_\theta(s)$ and $C_1(s)$

Giving results tableted below.

Table 1: Calculated and determined values of system behaviour

Feature	Value
K_p	19
K_d	4
Overshoot (%)	6.5631
Settling time (s)	2.2261
Rise time (s)	0.32165
Peak time (s)	0.63352
Steady-state gain (°)	0.97229

(b) Implementation and Nonlinear Simulation with Saturation Block:

Using the values found in part a, the inner loop can now be created and implemented in Simulink. The figure below shows the block diagram for the inner loop compensator.

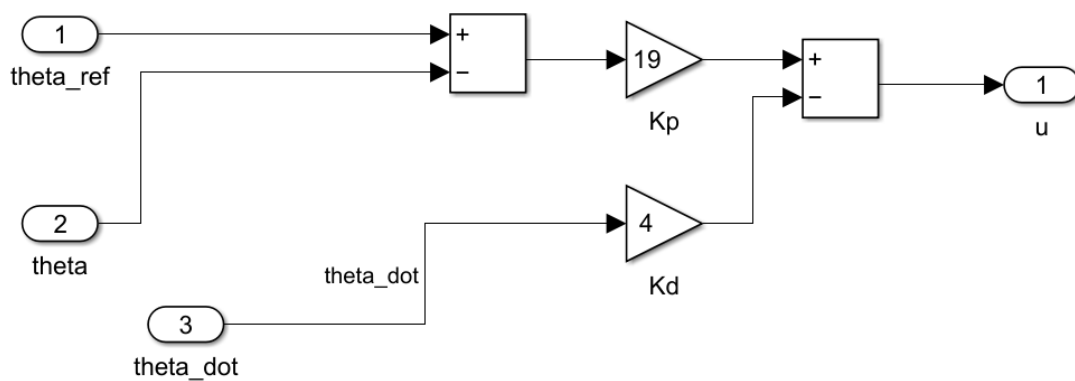


Figure 4: Inner loop compensator block diagram

Additionally, a saturation block was used between the compensator and the plant to prevent the gimbal angle from exceeding a magnitude of 15 degrees. The figures below show the plots of the rocket angle under the specified conditions.

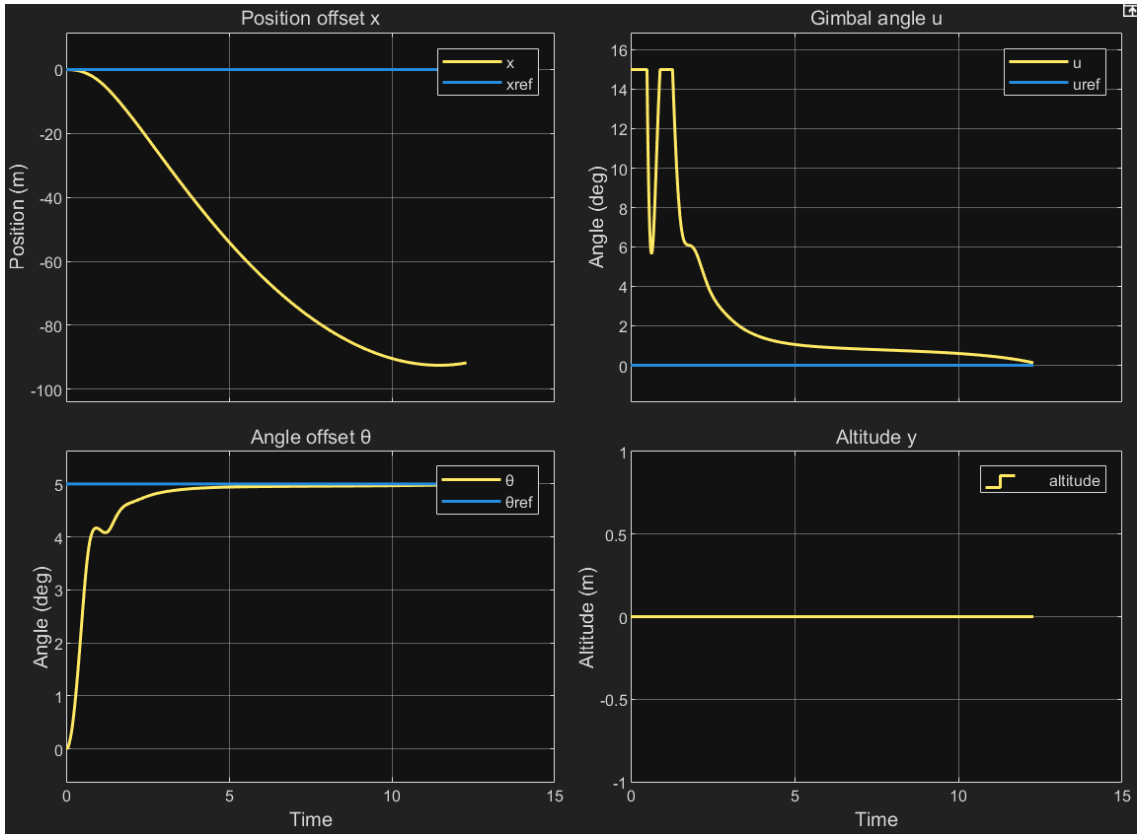


Figure 5: Inner loop implementation graphs (initial conditions: $\dot{y}_0 = -236 \text{ m/s}$, $\theta_{ref} = 5^\circ$)

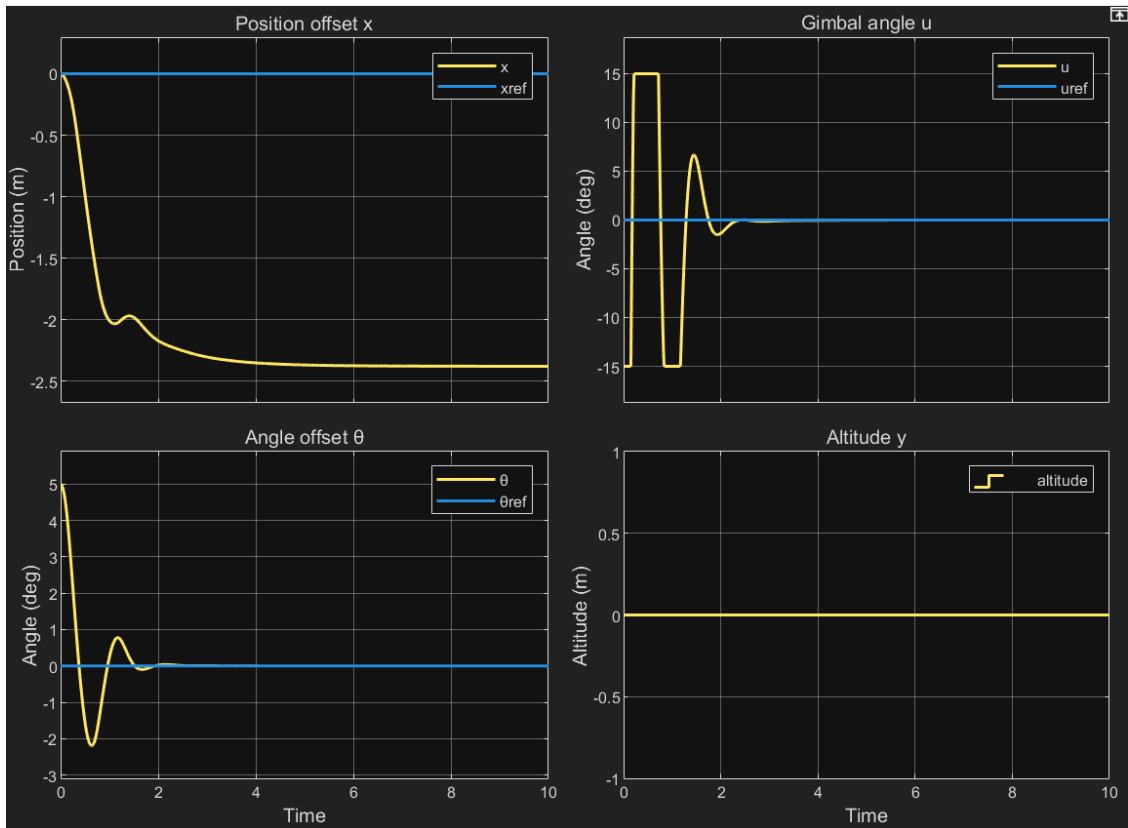


Figure 6: Inner loop implementation graphs (initial conditions: $\dot{y}_0 = -236 \text{ m/s}$, $\theta_{ref} = 0^\circ$, $\theta_0 = 5^\circ$)

The graphs show that the compensator works effectively. For both sets of initial conditions, both gimbal angle and angle offset reach the reference value at steady state. The effects of the saturation can be seen in the gimbal angle graphs, as the graphs level out, instead of exceeding, the magnitude of 15 degrees. In essence, the angle offset experiences a brief oscillation before levelling out at the reference point. The settling time requirement is also met, as can be shown where the graph nearly meets the reference point at $t = 2.5$ seconds.

(c) Outer loop controller design $C_2(s)$ for lateral position regulation:

Using the inner-loop controller designed in (a), the effective plant for the outer-loop design was computed in MATLAB (see Appendix) as shown below.

```

Inner closed-loop transfer function Ttheta(s) = theta/theta_ref
ans =

      30.59 (s+0.833)
      -----
      (s+0.7529) (s^2 + 6.106s + 34.81)

Continuous-time zero/pole/gain model.
Model Properties
Effective outer-loop plant Gin(s) = x/theta_ref
ans =

      -6.403 (s^2 + 5.2)
      -----
      s (s+0.7529) (s^2 + 6.106s + 34.81)

Continuous-time zero/pole/gain model.
Model Properties

```

Figure 7: Outer loop and inner closed loop factored transfer functions:

The inner closed-loop transfer function from $\theta_{\text{ref}}(s)$ to $\theta(s)$ is:

$$T_{\theta}(s) = \frac{\theta(s)}{\theta_{\text{ref}}(s)} = \frac{30.59(s + 0.833)}{(s + 0.7529)(s^2 + 6.106s + 34.81)} \quad (17)$$

The corresponding effective plant seen by the outer loop from $\theta_{\text{ref}}(s)$ to $x(s)$ is:

$$G_{\text{in}}(s) = \frac{X(s)}{\theta_{\text{ref}}(s)} = \frac{-6.403(s^2 + 5.2)}{s(s + 0.7529)(s^2 + 6.106s + 34.81)} \quad (18)$$

This is already in factored form, so the poles and zeros can be identified directly. The zeros are given by the roots of $(s^2 + 5.2 = 0)$, giving

$$s = \pm 2.28j \quad (19)$$

The poles are given by the denominator,

$$s = 0 \quad s = -0.7529 \quad (20)$$

together with the complex-conjugate poles from

$$s^2 + 6.106s + 34.81 = 0 \quad (21)$$

Therefore, the inner loop has been successfully closed and the resulting transfer function $G_{in}(s)$ can now be used as the plant for the design of the outer-loop compensator $C_2(s)$.

(d) Root Locus Design of the Outer-Loop Controller:

The effective plant obtained for the outer-loop design is

$$G_{in}(s) = \frac{-6.403(s^2 + 5.2)}{s(s + 0.7529)(s^2 + 6.106s + 34.81)} \quad (22)$$

With the root locus plot below, since $G_{in}(s)$ has an overall negative sign a stabilisation controller required a negative proportional gain. So, the controller should be designed using the root locus $-G_{in}(s)$, so this root locus plot will only be used this for observation of plant behaviour.

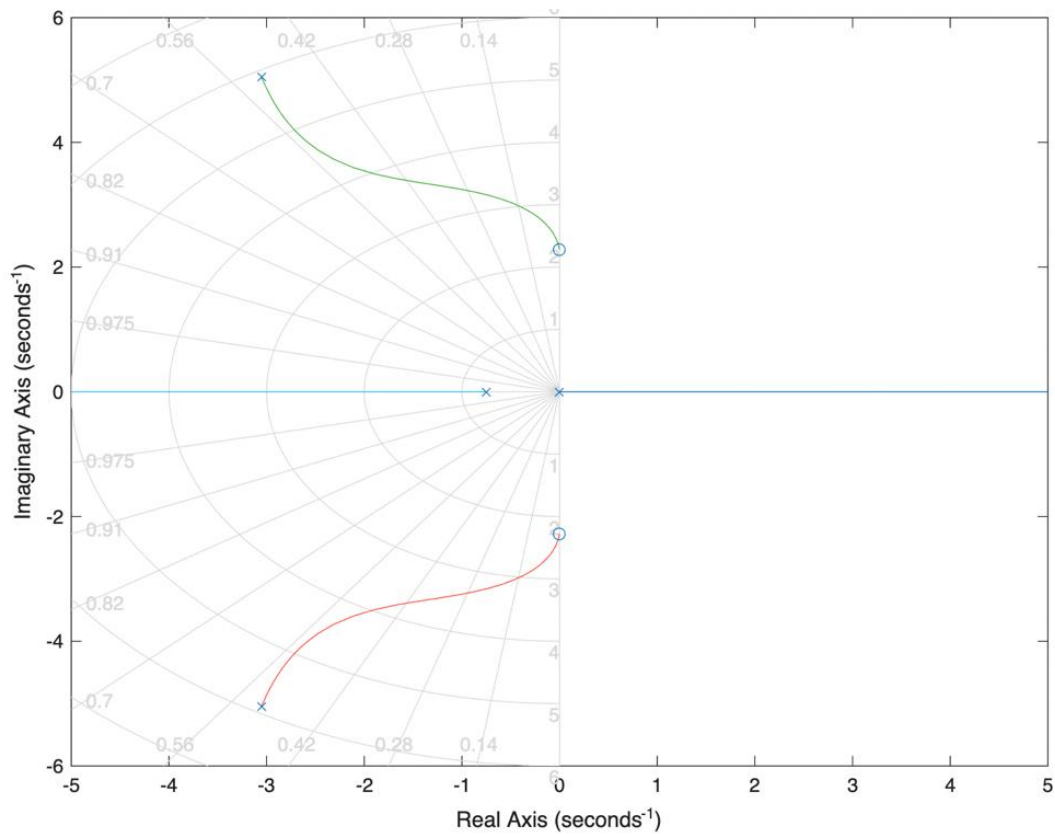


Figure 8: Root locus of $G_{in}(s)$

From inspection we can determine that a proportional controller is sufficient for lateral position control since $G_{in}(s)$ contains a pole at the origin, the plant is type 1. Therefore, with unity feedback, a proportional controller is sufficient to achieve zero steady-state error for a step reference input, provided the closed-loop system is stable.

Because $G_{in}(s)$ has an overall negative sign, a negative proportional gain is required for stabilisation. Therefore, the root locus should be plotted again for $-G_{in}(s)$, which is equivalent considering negative proportional gains in the controller.

To achieve a settling time of roughly 12 seconds, the dominant closed loop poles should be close to the line on the $-G_{in}(s)$ Root Locus plot below.

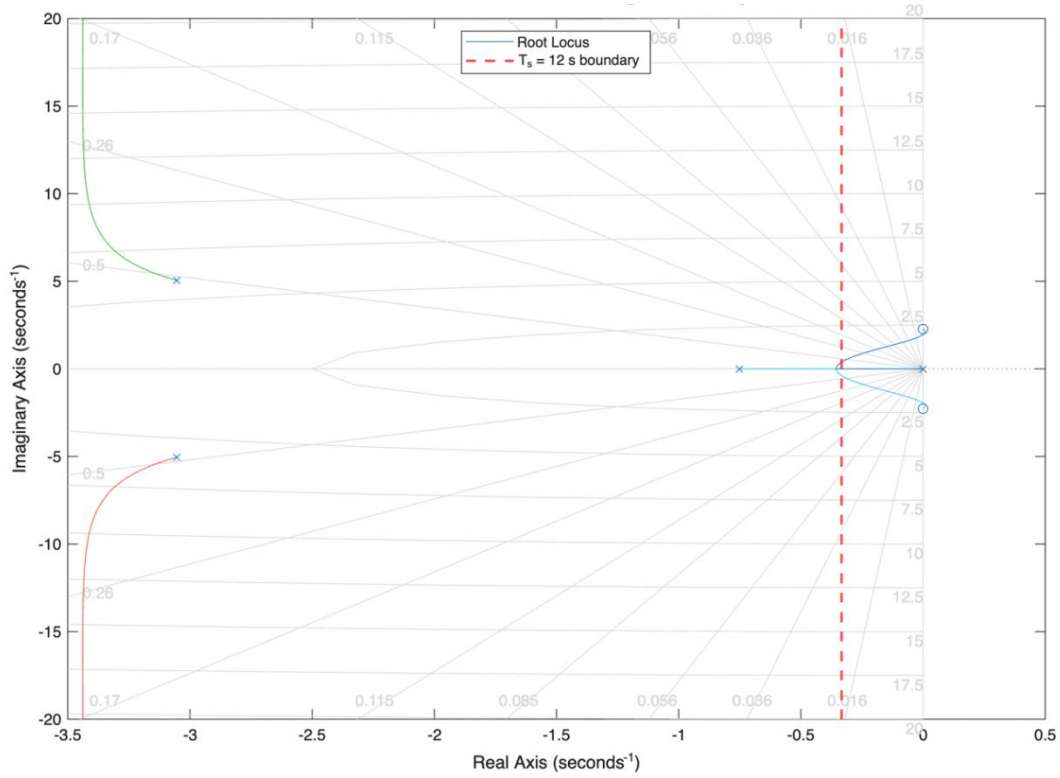


Figure 9: Root locus of $-G_{in}(s)$ including admissible region

From inspection of plot, the dominant closed-loop poles should have a real part approximately

$$\text{Re}(s) = -\frac{4}{12} = -0.333 \quad (23)$$

from,

$$\zeta \omega_n = \frac{4}{t_s}, \quad t_s = \frac{4}{\zeta \omega_n} \quad (24)$$

A suitable proportional gain is selected as

$$C_2(s) = K = -0.28 \quad (25)$$

This places the dominant poles approximately at

$$s = -0.34 \pm 0.34j \quad (26)$$

which gives an estimated settling time corresponding to approximately 12 seconds,

$$t_s = \frac{4}{0.34} \approx 11.76 \text{ s} \quad (27)$$

Verification in MATLAB using step response:

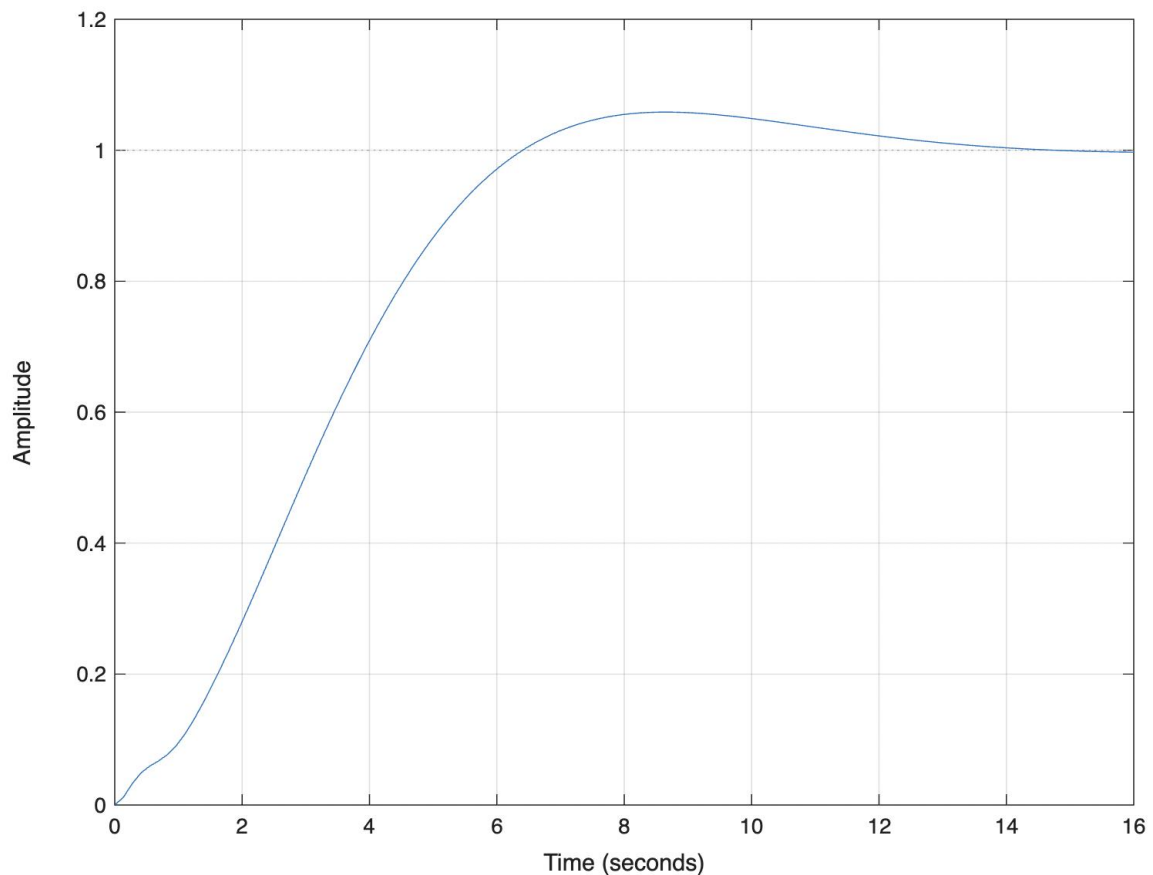


Figure 10: Closed loop step response for outer loop

With the following features, DC gain, and poles:

```
ans = struct with fields:
    RiseTime: 4.2326
    TransientTime: 12.1589
    SettlingTime: 12.1589
    SettlingMin: 0.9108
    SettlingMax: 1.0583
    Overshoot: 5.8291
    Undershoot: 0
    Peak: 1.0583
    PeakTime: 8.6665

ans = 1

ans = 4x1 complex
    -3.0947 + 5.2178i
    -3.0947 - 5.2178i
    -0.3348 + 0.3758i
    -0.3348 - 0.3758i
```

Figure 11: Closed loop system characteristics

The response is stable and settles in approximately 12 seconds, while also achieving zero steady-state error for a step input. This confirms that a proportional controller is sufficient for the outer-loop design.

(e) Simulation of completed controller design (with both loops) on the actual rocket in Simulink

Can the rocket land safely with the following initial conditions:

$\dot{y}_0 = -236 \text{ m/s}$, $x_0 = 50 \text{ m}$, Every other initial conditions zero. No wind.

Answer: yes!

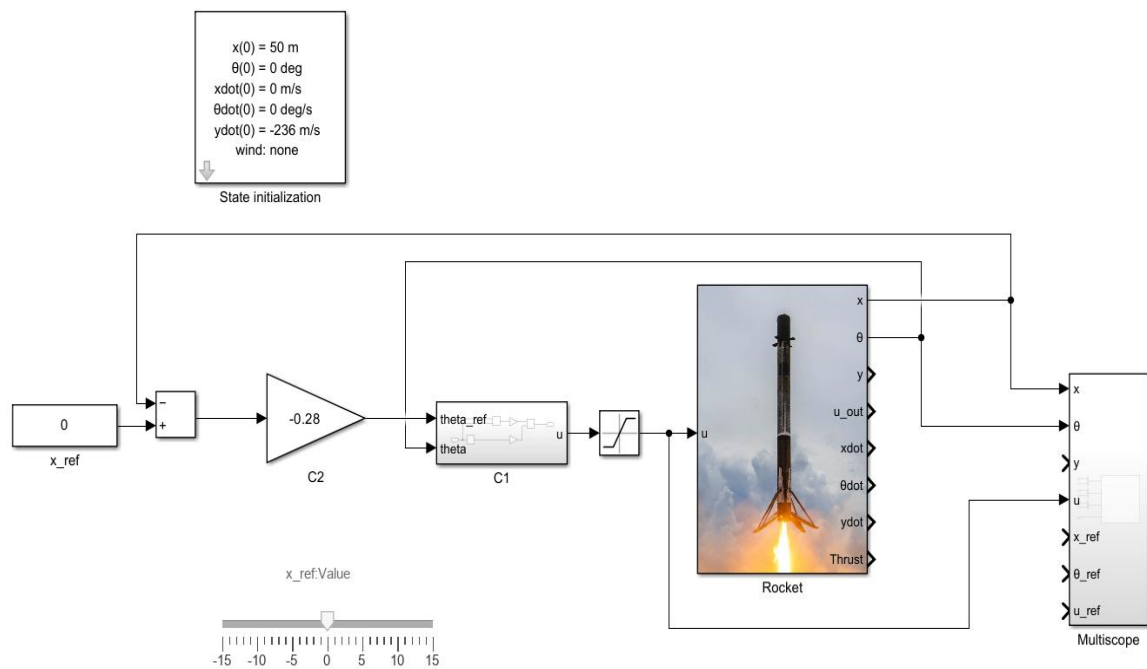


Figure 12: Final Simulink block diagram of closed loop system

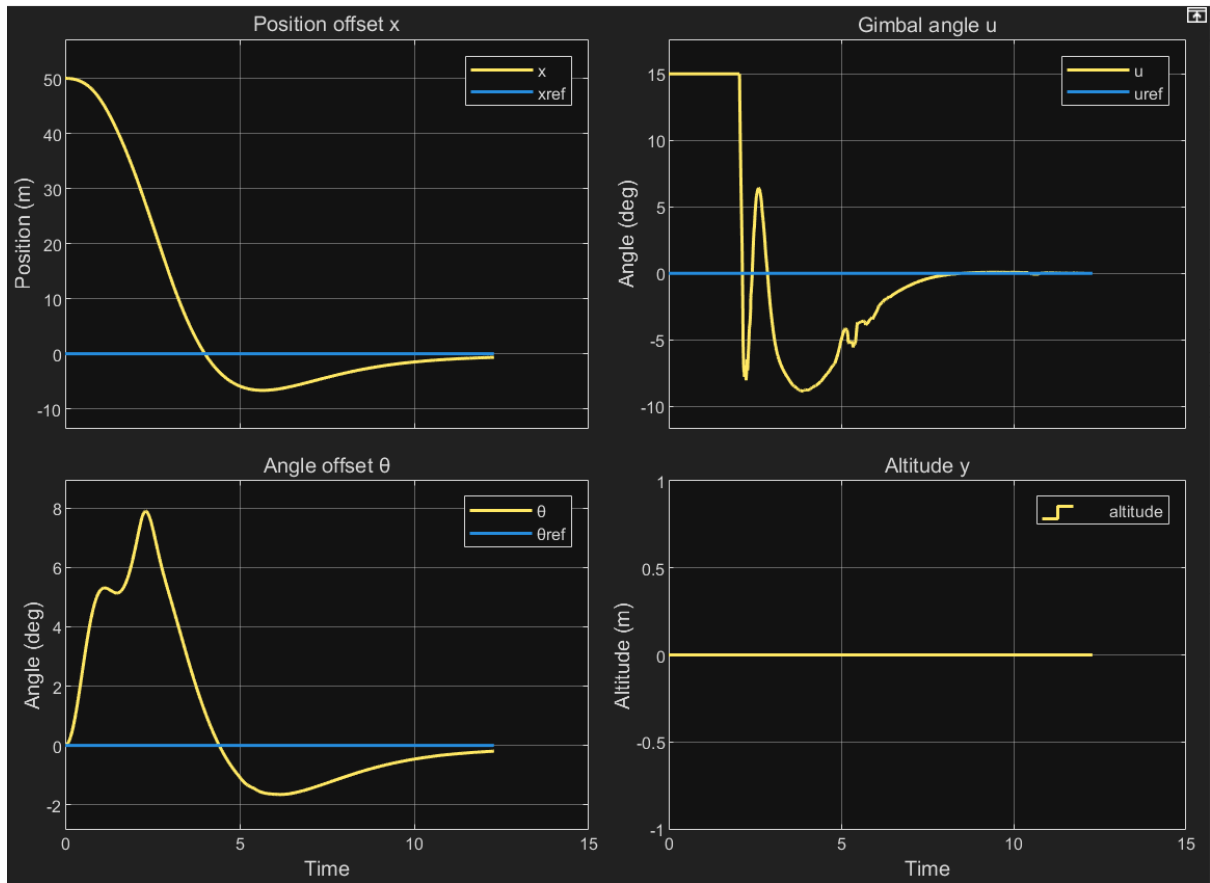


Figure 14: Graphs of closed loop system under specified conditions

```
*** SUCCESSFUL LANDING! ***
You landed safely on the pad!
```

Figure 15: Successful landing message

Appendix

```
s = tf('s');

G = 1.61*(s+0.833)/((s+0.213)*(s^2+0.206*s+3.41));

Kp = 19;
Kd = 4;

T = minreal((Kp*G)/(1 + (Kp + Kd*s)*G));

figure;
step(T);
grid on;
```

Figure 16: MATLAB code for Figure 3

```
s = tf('s');

Gtheta = 1.61*(s+0.833)/((s+0.213)*(s^2 + 0.206*s + 3.41));
Gx = -0.337*(s^2 + 5.20)/(s*(s+0.213)*(s^2 + 0.206*s + 3.41));

Kp = 19;
Kd = 4;

% Inner closed-loop transfer function, theta_ref to theta
Ttheta = minreal((Kp*Gtheta)/(1 + (Kp + Kd*s)*Gtheta));

% Effective plant for outer loop, theta_ref to x
Gin = minreal((Kp*Gx)/(1 + (Kp + Kd*s)*Gtheta));

disp('Inner closed-loop transfer function Ttheta(s) = theta/theta_ref')
zpk(Ttheta)

disp('Effective outer-loop plant Gin(s) = x/theta_ref')
zpk(Gin)
```

Figure 17: MATLAB code for Figure 7

```
s = tf('s');

Gin = -6.403*(s^2 + 5.2)/(s*(s+0.7529)*(s^2 + 6.106*s + 34.81));

K = -0.28;
```

```
T = feedback(K*Gin,1);

figure
step(T)
grid on
title('Closed-loop step response for outer loop')

stepinfo(T)
dcgain(T)
pole(T)
```

Figure 18: MATLAB code for Figure 10 & Figure 11